Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Compute Grad-CAM
2. Abbreviated Title	Compute Grad-CAM
3. Use Case Id	UC-12
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine computes Gradient-weighted Class Activation Mapping (Grad-CAM) on the CNN branch's final convolutional layer to produce a heatmap highlighting which regions of the GAF image most influenced the prediction.
5.1. Pre Conditions	1. Model inference (UC-11) has been completed. 2. Gradients are available (model in eval mode with gradient hooks).
5.2. Task Sequence	1. System registers gradient hooks on the target convolutional layer. 2. System performs a forward pass with the patient input. 3. System computes gradients of predicted class w.r.t. target activations. 4. System performs global average pooling to get channel weights. 5. System computes weighted sum of activation maps and applies ReLU. 6. System maps heatmap regions back to clinical feature names.
5.3. Post Conditions	1. A 2D Grad-CAM heatmap overlay is generated. 2. Heatmap is mapped to feature names for clinical interpretation.
6. Modification History	Date: September 11, 2026
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